YAHYA MENKARI

AI & MACHINE LEARNING ENGINEER

RAG • Agentic & Workflow Automation • Time-Series Forecasting

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PROFESSIONAL SUMMARY

AI & Machine Learning Engineer with hands-on experience building end-to-end ML, RAG, and automation systems. Developed a 24-hour weather forecasting pipeline using 43,000+ real-world observations and built AI applications combining graph/vector retrieval, multi-agent workflows, and cloud deployment. Experienced in taking projects from data processing and modeling through workflow orchestration, user-facing applications, and deployment.

PROFESSIONAL EXPERIENCE

Machine Learning Research Intern | Technical University of Sofia, Bulgaria [View in Github]| Feb 2026 – Jun 2026

- Developed an end-to-end forecasting pipeline using **43,000+ hourly observations** collected over five years.
- Engineered **43 predictive features** using cyclical encodings, lag variables, and rolling statistics.
- Benchmarked **XGBoost, GRU, CNN-LSTM, CNN-GRU, and TCN** for 24-hour forecasting; **optimized XGBoost** achieved **MAE 2.34, RMSE 3.05, and R^2 0.91** on unseen data.

AI Research Intern | Archipel Cognitive, Morocco | Jul 2024 – Sep 2024

- Researched and designed **RAG architectures for social-media content retrieval**, covering ingestion, embeddings, vector databases, semantic retrieval, and LLM integration.
- Developed a **multi-stage X/Twitter content extraction workflow** and investigated **CDC-based streaming retrieval** for incremental vector updates and **Agentic RAG**

PROJECTS

AI-Powered Job Discovery & Outreach Automation Platform | *n8n | Python | Streamlit | Docker | GCP*

- Designed and deployed a multi-stage automation platform for job discovery, qualification, contact research, outreach preparation, and application tracking.
- Built modular n8n workflows with conditional routing, deduplication, persistent state management, and a Streamlit dashboard deployed in Docker on GCP.

Graph-Based AI Symptom Checker (*View GitHub*)| Neo4j, Qdrant, RAG, PubMed

- Built a knowledge-graph and semantic retrieval system combining Neo4j, Qdrant, PubMed, and RAG.

Medical AI Assistant with Specialist Referral System(*View GitHub*) | CrewAI, Gemini, Streamlit

- Developed a multi-agent healthcare assistant automating specialist referral, report generation, and information retrieval workflows.

TECHNICAL SKILLS

- **Programming**: Python, SQL, Java, C
- **AI & Machine Learning**: Scikit-learn, XGBoost, Deep Learning, Time-Series Forecasting, RAG, Agentic AI, LLM Orchestration
- **Automation & Applications**: n8n, Workflow Automation, Streamlit, REST APIs
- **Data & Retrieval**: MySQL, MongoDB, Neo4j, Qdrant, Semantic Search, Knowledge Graphs
- **Cloud & DevOps**: Docker, GCP, Azure, Git
- **Analytics**: Power BI, SAP Analytics Cloud

EDUCATION

International University of Rabat (UIR), Morocco

- M.Eng. in Computer Science – Big Data & Artificial Intelligence (2021 – 2026) – Mention Très Bien

Technical University of Sofia, Bulgaria

- Exchange Program – IT for Business Management (2025 – 2026)

Languages:

Arabic — Native | French — Bilingual | English — B2+